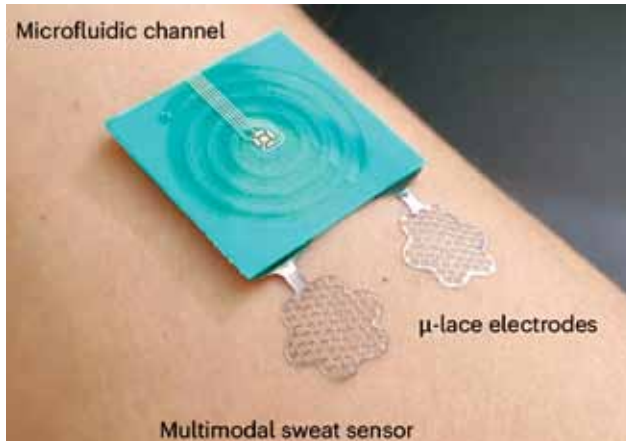




Real-Time Tracking Of Hydration And Stress

UC Berkeley (California, USA) researchers have developed a thin, breathable wearable sensor that tracks electrodermal activity (EDA) to monitor real-time hydration and stress levels. Unlike bulky alternatives, this lightweight device fits easily into fitness trackers without disposable parts. It benefits athletes, students, and office workers by



The multimodal sweat sensor, placed on the skin, features a microfluidic channel and breathable μ -lace electrodes. The device's compact form factor and ability to conform closely to the skin enable comfortable and stable wear (Image: Courtesy of the researchers)

promoting better hydration and focus. With sweat-permeable electrodes that boost comfort and accuracy, the team now aims to personalise the technology based on environment and individual skin types.

Robots Get Unique 'Fingerprints'

Scientists at UNIST (Ulsan, South Korea) have developed a flexible electronic skin that mimics—and surpasses—the uniqueness of human fingerprints. Created using a polymer called SEBS and treated with solvent, the skin forms random wrinkle patterns, with only a 1 in 10^{43} chance of duplication in a 1mm^2 area, making it vastly more unique than human prints. Durable against heat, impact, and moisture, the skin gives robots enhanced sensory capabilities. In tests, it enabled a robotic hand to grip, detect textures, and even react to heat like a human reflex.

UAV Navigation In GPS-Denied Environments

A study from Prince Sultan University (Riyadh, Saudi Arabia) tackles the challenge of UAV navigation without reliable GPS, which is common in urban canyons or disaster zones. It explores hybrid systems that combine sensors like

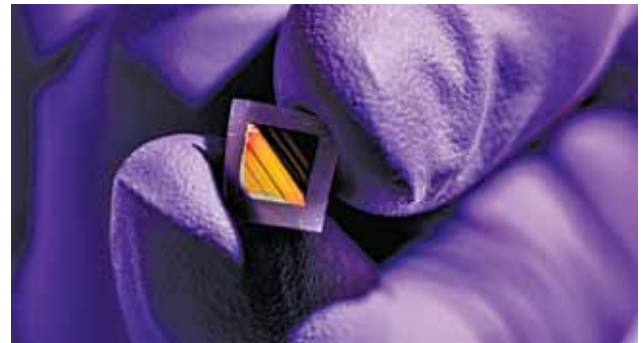
lidar, radar, and IMUs for accurate real-time localisation. The research reviews over 130 studies, comparing absolute methods using pre-mapped data and relative methods like SLAM and visual-inertial odometry. Vision-based systems and AI offer promise but face limitations. Lead author Dr Imen Jarraya emphasises the need for optimised sensor fusion to enable reliable UAV operations across diverse, GPS-denied environments.

Hybrid Material Doubles Thermoelectric Efficiency

Team at Los Alamos National Laboratory (USA), has developed a hybrid thermoelectric material that significantly enhances energy conversion efficiency by separating heat and electrical transport. The material combines an iron-based alloy with a bismuth-antimony mix, compressed to form a compact structure while maintaining distinct atomic phases. This design disrupts heat transfer due to mismatched atomic vibrations but enables efficient charge movement. The bismuth-antimony acts as a topological insulator, conducting electricity along surfaces with minimal resistance. This innovation could greatly improve energy harvesting in compact, autonomous IoT devices.

Thin Film Enables Night Vision

MIT (USA) engineers have developed a way to grow and lift off ultra-thin 'skins' of electronic materials, enabling devices like sensors and imaging systems. They created a 10-nanometre pyroelectric film that detects heat and radiation in the



The newly developed film could enable lighter, more portable, and highly accurate far-infrared (IR) sensing devices, with potential applications for night-vision eyewear and autonomous driving in foggy conditions (Credit: Adam Glanzman)

far-infrared range. This film may allow infrared sensors that don't need cooling, which is helpful for night-vision eyewear and vehicle sensors. The lift-off method, which works even without a graphene layer, allows the reuse of the base material. Researchers are now building a full night-vision system and exploring its use in other electronic applications.